

LESSON 4.7

APPLYING EXPONENT LAWS TO ALGEBRAIC EXPRESSIONS – PART 1



LESSON INTRODUCTION

MA.8.AR.1.1 Apply laws of exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

LEARNING TARGETS

- I can write an equivalent algebraic expression by applying the product of powers, power of a power, or power of a product laws of exponents.

BENCHMARK COHERENCE

BUILDING ON

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational bases.

WORKING TOWARD

MA.9.12.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

ESSENTIAL QUESTIONS

- Can the power of a power, product of powers, and power of a product exponent laws be used for expressions with variable bases?
- How do you apply the laws of exponents to write equivalent algebraic expressions?

LESSON NARRATIVE

In this lesson, students build from their previous understanding of applying the laws of exponents to include generating equivalent algebraic expressions using exponent laws. Students explore and discover the power of a power, product of powers, and power of a product exponent laws apply to expressions with numerical and variable bases when generating equivalent algebraic expressions.

COMPONENT SUMMARY

4.7.1 WARM-UP (~5 MIN.)

Draw the next step in a pattern with hexagons and squares, then determine if a given expression represents the pattern

4.7.3 EXPLORATION (15 MIN.)

Use the laws of exponents to rewrite algebraic expressions

4.7.5 ACTIVITY (10 MIN.)

Identify the missing value to make each equation a true, equivalent statement

4.7.2 GUIDED INSTRUCTION (10 MIN.)

Connecting equivalent algebraic expressions by using substitution to compare values

4.7.4 BRING IT TOGETHER (5-10 MIN.)

Applying laws of exponents to numerical and variable bases

4.7.6 WRAP-UP (~5 MIN.)

Apply the laws of exponents to generate equivalent algebraic expressions

RESOURCES

GLOSSARY

Key Terms:

- Power of a power
- Product of powers
- Power of a product

Additional Terms:

- N/A

MATERIALS

- Desmos scientific calculators
- (Optional) Highlighters or colored pencils
- (Optional) Pattern blocks

ADDITIONAL PREPARATION

- Consider displaying an anchor chart of the laws of exponents or allow students to use their reference sheet (or foldable) that was suggested in Lesson 1.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the power of a power property as the distributive property.
- When applying the product of powers law, students may add the coefficients rather than multiply them.
- When applying the power of a product law, students may multiply the power by the coefficient rather than raising the coefficient to that power.

THINGS TO CONSIDER

- Help students connect working with algebraic expressions to their knowledge from Lessons 1-6 of applying exponent laws to evaluate numerical expressions and to generate equivalent numerical expressions.
- For additional scaffolding if needed, encourage students to write expressions in expanded form throughout the lesson.
- Students should be allowed to use a calculator on question 1c, 1d, and 1e in 4.7.2 Guided Instruction.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

During 4.7.4 Bring It Together, consider engaging students in a discussion about how applying the laws of exponents to algebraic expressions compares and connects to applying the laws of exponents to numerical expressions. Encourage students to use an example from a previous lesson in the unit and an example from the current lesson to help explain the connection.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

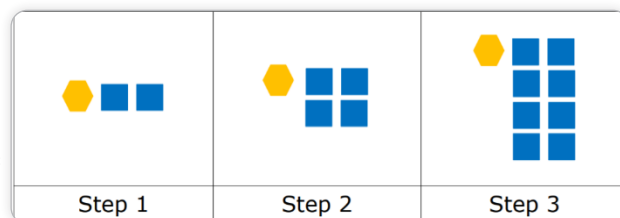
During 4.7.3 Exploration, students explore several algebraic expressions written in different ways and the relationships between the coefficients and variables of the given expressions.

DIFFERENTIATE INSTRUCTION

Provide a kinesthetic entry point by providing students with pattern blocks to model the pattern in 4.7.1 Warm-Up.

4.7.3 Exploration and 4.7.4 Bring It Together provide opportunities for visual entry points by having students highlight, circle, or underline the pieces of each expression that can be combined. For example: $2z^4 \cdot 5z^3 = 2 \cdot 5 \cdot z^4 \cdot z^3 = 10 \cdot z^{4+3} = 10z^7$.

Additional differentiation opportunities are denoted throughout the lesson guide.



4.7.1 WARM-UP: PATTERNS

DIFFERENTIATE INSTRUCTION

Provide a kinesthetic entry point by providing students with pattern blocks to model the pattern.

TEACHER MOVES

It's not the expectation that every student is able to see this, but many will be able to make the connections. It's a purposeful extension question.

ENRICHMENT

If time permits, consider asking students which value in Ibrahim's expression corresponds with each shape in the pattern and to discuss with a partner how they know.

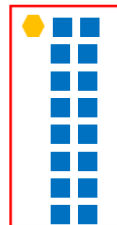


4.7.1 Warm-Up: Patterns

- The first three steps in a pattern are shown in the table. The pattern is created using yellow hexagons and blue squares.

Step 1	Step 2	Step 3

- Draw what you think Step 4 in the pattern will look like.



- Ibrahim said the pattern could be represented by the expression $1^n + 2^n$, where n is the number of the step in the pattern. Explain whether or not you agree with Ibrahim.

Answers will vary. I agree with Ibrahim because the number of hexagons is always 1, and 1 to any power is always 1. The number of squares equals 2 to the power of n , where n is the number of the step.



4.7.2 Guided Instruction: Equivalent Algebraic Expressions

Laws of exponents can be applied to generate equivalent algebraic expressions that are similar to numerical expressions.

1. A square is shown, where each side has a length represented by the expression $3a^5$.



The formula $A = s^2$ can be used to find the area of the square, where s is the side length of the square.

- a. Using this formula, the area of the square shown can be expressed as $(3a^5)^2$. Discuss with your partner how this expression represents the area of the square.
Answers will vary. The length of a side, $3a^5$, is substituted for s and then squared.
- b. Complete the steps to find an equivalent expression that represents the area of the square.

$$\begin{aligned} &(3a^5)^2 \\ &= 3a^5 \cdot 3a^5 \\ &= 3 \cdot 3 \cdot a^5 \cdot a^5 \\ &= 3 \cdot 3 \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \\ &= 9a^{10} \end{aligned}$$

- c. Suppose $a = 1.5$ units. Find the length of one side of the square using a calculator.
22.78125 units
- d. Use your answer to Part C to find the area of the square. Round to the nearest thousandth.
518.985 units²
- e. Use the expression generated in Part B to find the area of the square. Round to the nearest thousandth.
518.985 units²
- f. What do you notice?
The two values for the area of the square are the same.

4.7.2 GUIDED INSTRUCTION: EQUIVALENT ALGEBRAIC EXPRESSIONS

ENRICHMENT

How did squaring the algebraic expression impact the coefficient? How did squaring the algebraic expression impact the exponent?

TEACHER MOVES

Allow students to use a calculator to evaluate questions 1c, 1d, and 1e. Likely in the first six lessons, they would have been shown how to enter an exponential expression into whatever calculator they use. If not, they will need to be shown this here.

TEACHER MOVES

In problems that require several steps, students should not round until the final step. Rounding mid-step could significantly change the answer. In question 1c, students should not round their answer. For questions 1d and 1e, rounding to the thousandth is intentional.

4.7.3 EXPLORATION: LAWS OF EXPONENTS WITH ALGEBRAIC EXPRESSIONS

DIFFERENTIATE INSTRUCTION

Consider allowing students to use colored pencils or highlighters to write each expression in expanded form. The partially completed first row shows coloring of each expression as it's expanded. Another visual representation for students would be to circle or underline the components of each expression that can be combined. For example, in the partially completed row, students would circle all of the c 's, since those can be combined. In the second row, when students write out the expanded form, they would underline the 3 and 2 and circle all of the z 's.

TEACHER MOVES

Students may make errors during the Exploration, including not multiplying the coefficients or applying the laws of exponents incorrectly when there are variable bases. Allow the opportunity for them to correct their own thinking as they work through and discuss with their partner. Do not be afraid to let the mistakes happen; discuss them during the transition to the next component.

ENRICHMENT

How does applying exponent laws to algebraic expressions compare to applying exponent laws to numeric expressions?



4.7.3 Exploration: Laws of Exponents with Algebraic Expressions

- Several algebraic expressions are given in the table.
 - For each expression, work with your partner to re-write the expression in expanded form. Then, write an equivalent algebraic expression with one variable base and one coefficient. The first row has been partially completed.
 - Discuss with your partner what you notice about how the coefficients and variable powers of the equivalent expressions compare to the given expressions.
 - Gavin said properties of exponents could be applied to the original expressions to generate the same equivalent expression. Match each pair of expressions with the exponent law that applies. Then, complete the explanations.

Given Expression	Expanded Form	Equivalent Expression
$5c^2 \cdot c^6$	$5 \cdot c \cdot c \cdot c \cdot c \cdot c \cdot c \cdot c$	$5c^8$
$3z^5 \cdot 2z$	$3 \cdot 2 \cdot z \cdot z \cdot z \cdot z \cdot z$	$6z^6$
$(2y^6)^2$	$(2 \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y)(2 \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y)$	$4y^{12}$
$(m^2)^4$	$(m \cdot m)(m \cdot m)(m \cdot m)(m \cdot m)$	m^8
$(x \cdot y)^3$	$(x \cdot y)(x \cdot y)(x \cdot y)$	$x^3 \cdot y^3$
$(3c \cdot d)^2$	$(3c \cdot d)(3c \cdot d)$	$9c^2 \cdot d^2$

Original Expressions	Which property of exponents applies?	Explanation
$5c^2 \cdot c^6$ and $3z^5 \cdot 2z$	☒ product of powers ☐ power of a product ☐ power of a power	The powers of the variable bases from the original expression were ☒ added to ☐ multiplied result in the power of the variable base in the equivalent expression.
$(2y^6)^2$ and $(m^2)^4$	☐ product of powers ☐ power of a product ☒ power of a power	The powers of the variable bases from the original expression were ☐ added to ☒ multiplied result in the power of the variable base in the equivalent expression.
$(x \cdot y)^3$ and $(3c \cdot d)^2$	☐ product of powers ☒ power of a product ☐ power of a power	Each factor in the expression within parentheses was raised to the ☒ power of the product. ☐ power of the power.

- d. Use words from the bank to complete the explanations of how the laws of exponents applied to the coefficients when generating the equivalent expressions.

multiplied	power	product	same
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In the expressions where product of powers law applied to the variable factors, the coefficients were multiplied to find the coefficient of the product.

In the expressions where the power of a power law applied, the coefficient was also raised to the power of the power.

In the expressions where the power of a product law applied, the coefficients of the factors were also raised to the same power.

4.7.3 EXPLORATION: LAWS OF EXPONENTS WITH ALGEBRAIC EXPRESSIONS (CONTINUED)

TEACHER MOVES

Consider allowing students to share and discuss their responses to Question 1d with a partner.

4.7.4 BRING IT TOGETHER: APPLYING LAWS OF EXPONENTS TO VARIABLES

TEACHER MOVES

Compare and Connect

Consider starting the component by engaging students in a partner discussion to compare and connect their understanding of applying the product of powers, power of a power, and power of a product laws to expressions with variable bases. Then, have students add in, explaining their understanding when an expression has a variable and numerical base.

DIFFERENTIATE INSTRUCTION

Consider allowing students to use colored pencils or highlighters to identify the parts of each expression that will be combined before writing the equivalent expression.



4.7.4 Bring It Together: Applying Laws of Exponents to Variables



The **product of powers**, **power of a power**, and **power of a product** laws of exponents apply to both variable and numerical bases when generating equivalent algebraic expressions.

1. Apply the law of exponents to generate equivalent expressions.

Product of Powers	Power of a Power	Power of a Product
$a^m \cdot a^n = a^{m+n}$	$(a^m)^n = a^{mn}$	$(ab)^m = a^m \cdot b^m$
$x^7 \cdot x^9 = x^{16}$	$(g^2)^3 = g^6$	$(x^2y)^9 = x^{18} \cdot y^9$
$2z^4 \cdot 5z^3 = 10z^7$	$(h^0)^6 = h^0 = 1$	$(4p^3)^3 = 64p^9$



4.7.5 Activity: Missing Numbers

- Work with your partner to use the digits 0 – 9 to complete the equations. Each digit is used exactly once. Blue boxes are coefficients and orange boxes are exponents.

0 1 2 3 4 5 6 7 8 9

a. $k^{\boxed{5}} \cdot k^2 = k^7$

b. $(g^{\boxed{2}})^{\boxed{3}} = g^7$
The exponents 1 and 7 may be used in either place.

c. $(\boxed{2} \cdot x^4)^{\boxed{3}} = 8x^{12}$

d. $r^{\boxed{0}} \cdot r^5 \cdot r^3 = r^{\boxed{8}}$

e. $(3c^{\boxed{4}} \cdot b^3)^2 = \boxed{9} \cdot c^8 \cdot b^{\boxed{6}}$

4.7.5 ACTIVITY: MISSING NUMBERS

TEACHER MOVES

Prompt students to consider whether more than one solution is possible. Have a variety of students share their work and justifications.

ENRICHMENT

How did you determine the coefficients? How did you determine the powers of the variables? Which law of exponents did you use?



4.7.6 Wrap-Up: Ticket Out the Door

- Apply the laws of exponents to generate an equivalent algebraic expression for each of the following.
 - $(5y)^2 = 5y \cdot 5y = 25y^2$
 - $x^6 \cdot x^3 = x^9$
 - $(t^5)^2 = t^5 \cdot t^5 = t^{10}$

4.7.6 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

Proficiency on the 4.7.6 Wrap-Up includes: • fluency with identifying and applying the laws of exponents to generate equivalent algebraic expressions • fluency with adding and multiplying numbers

DIFFERENTIATE INSTRUCTION

Student responses, coupled with what was observed during instruction, can be a formative assessment to provide information for differentiating instruction in later lessons.